Lecture 04 — Inner product

Mon Aug 31

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Boyd video (optional): Lecture 5 — inner product · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 1.4 (inner product); Boyd 1.1 stack and slice (Quiz 1 prep).

Learning objectives

By the end of class, students should be able to:

1. Compute a dot product.
2. Interpret it as a weighted sum.
3. Interpret its sign and magnitude as alignment.
4. Recognize orthogonal vectors from a zero dot product.

Fill in — stack and slice

A **stacked** (or **block**) vector lists the entries of one vector after another. If b is an m-vector and c is an n-vector, then

$$(b, c) = (b_1, \dots, b_m,\ c_1, \dots, c_n)$$

is an $(m+n)$-vector. A scalar can be stacked with a vector: if a is a 3-vector, then $(1, a)$ is a _____-vector with first entry 1, then a_1, a_2, a_3.

Stacking is **not** vector addition: (a, b) has length $m+n$, not m.

A **slice** (or **subvector**) takes a contiguous block of entries. Boyd's notation $x_{r:s}$ means the vector of entries $x_r, x_{r+1}, \dots, x_s$. It has length _____. The endpoints are **inclusive**.

Example: if $z = (1, -1, 2, 0)$, then $z_{2:3} =$ _____.

If a and b are 10-vectors and c is a 20-vector, then $(a, b, c_{3:13})$ has length _____.

Fill in

For n-vectors a and b, the **inner product** (or **dot product**) is the scalar

$$a^\mathsf{T} b = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n = \sum_{i=1}^{n} _____.$$

It is only defined when a and b have the same _____.

If w is a weight vector and x is a data vector (same length), then $w^\mathsf{T} x$ is a _____ of the entries of x, using the weights in w.

Geometrically in $\mathbb{R}^2$ (and similarly in $\mathbb{R}^3$): if $a^\mathsf{T} b > 0$, the arrows point roughly the _____ direction; if $a^\mathsf{T} b < 0$, they point roughly _____ directions; if $a^\mathsf{T} b = 0$, the vectors are _____.

Two vectors with $a^\mathsf{T} b = 0$ are **orthogonal** (perpendicular). The inner product is commutative: $a^\mathsf{T} b =$ _____.

The quantity $a^\mathsf{T} a = a_1^2 + \cdots + a_n^2$ is always _____ (never negative).

Worked in class

W1. Let $b = (2, -1)$ and $c = (0, 3, 1)$. Write the stacked vector (b, c) explicitly. What is its length?

W2. Let $z = (1, -1, 2, 0)$. Write $z_{2:3}$ and $z_{1:2}$. If $x = (63, 59, 61, 66)$ is four daily high temperatures (°F), write the 3-vector of day-to-day **changes** as $x_{2:4} - x_{1:3}$.

W3. Compute $u^\mathsf{T} v$ for $u = (1, 1)$ and $v = (1, -1)$. Sketch both arrows in $\mathbb{R}^2$ from the origin. How does the zero dot product match the picture?

W4. Three quiz parts have scores (out of 1):

$$g = (0.80,\ 0.90,\ 0.70),$$

with weights

$$w = (0.50,\ 0.30,\ 0.20).$$

Compute the weighted total $w^{\mathsf{T}} g$. (Same idea as a weighted average when the weights sum to 1.)

W5. Morning wind displacements (east, north), in km/h:

$$a = (3, 1), \qquad b = (-1, 3).$$

Compute $a^{\mathsf{T}} b$. Is it positive, negative, or zero? In one sentence, what does the sign tell you about how the two directions line up?

Your turn

Y1. Let $a = (4, 1, 2)$ and $b = (1, 0, 1)$ be shopping lists (apples, bread, milk). Write (a, b) and state its length.

Y2. Let $c = (10, 12, 11, 14, 13)$ be five daily counts. Find $c_{2:4}$.

Y3. Compute $(2,-1)^{\mathsf{T}}(3,4)$ and $(1,0)^{\mathsf{T}}(0,1)$.

Y4. A café tracks daily sales in the order (espresso, milk, syrup):

$$x = (40,\ 30,\ 10),$$

with prices in dollars

$$p = (3,\ 2,\ 1).$$

What quantity does $p^{\mathsf{T}}x$ represent? Compute it.

Y5. Two shopping lists, in the order (apples, bread, milk):

$$r = (2,\ 1,\ 3), \qquad s = (4,\ 2,\ 6).$$

Compute $r^{\mathsf{T}}s$. Notice anything about r and s? If $t = (1, 0, -1)$, compute $r^{\mathsf{T}}t$ and say whether r and t are orthogonal.

Answers (Your turn)

Y1. $(a, b) = (4, 1, 2, 1, 0, 1)$; length 6.

Y2. $c_{2:4} = (12, 11, 14)$.

Y3. $(2, -1)^\mathsf{T}(3, 4) = 2 \cdot 3 + (-1) \cdot 4 = 2$. $(1, 0)^\mathsf{T}(0, 1) = 0$ (orthogonal).

Y4. Total revenue in dollars: $p^\mathsf{T}x = 3 \cdot 40 + 2 \cdot 30 + 1 \cdot 10 = 120 + 60 + 10 = 190$ dollars.

Y5. $r^\mathsf{T}s = 2 \cdot 4 + 1 \cdot 2 + 3 \cdot 6 = 8 + 2 + 18 = 28$. Also $s = 2r$, so the lists are proportional (same "direction" in data space). $r^\mathsf{T}t = 2 \cdot 1 + 1 \cdot 0 + 3 \cdot (-1) = 2 - 3 = -1 \neq 0$, so r and t are not orthogonal.